**MERU UNIVERSITY OF SCIENCE AND TECHNOLOGY**

**CCS 3251: WIRELESS AND MOBILE COMPUTING (45 CONTACT HOURS)**

**COURSE OUTLINE**

**Prerequisite:** Network System Design and Implementation

**Course Purpose**

The course examines mobile networks and relevant standards, highlighting issues unique to the mobile Computing environment and exploring the differences between conventional and mobile applications. Students will be introduced to and use mobile application frameworks and development environments to reinforce concepts covered in lectures.

**Expected Learning Outcomes**

By the end of this course the student should be able to:

i. Explain basics of Wireless voice and data communications technologies.
**ii.** Describe the working principles of wireless LAN and its standards.
**iii.** Demonstrate Wireless Application Protocols. Course Description

**Course Description**

**Week 1-2: Introduction to Wireless and Mobile Computing**

- Overview of wireless and mobile computing
- Definition of terms (Wireless, Mobile, Ubiquitous Computing)
- Historical perspective and evolution

**Week 3-4: Wireless and Mobile Network Architecture**

- Understanding mobile devices (PDA, laptops, web phones)
- Mobile operating systems
- IEEE 802.11 standard for wireless communication

**Week 5-6: Technologies for Wireless Communication**

- Satellite communication
- Microwave communication
- Infrared communication
- Bluetooth technology
- Zigbee protocol

**Week 7: Cellular Networks**

- Cell concept and cell types
- Handover strategies in cellular networks
- Frequency sharing in cellular networks
- Overview of GSM and GPRS technologies

**Week 8-9: Evolution of Mobile Networks**

- 1G to 4G technologies
- Advancements and features of each generation

**Week 10-11: Wireless Networks**

- Personal Area Networks (PANs)
- Wireless Local Area Networks (WLANs)
- Wireless Metropolitan Area Networks (WMANs)
- Wireless Wide Area Networks (WWANs)

**Week 12: Protocols for Wireless Networks**

- Mobile IP Architecture
- IPv4 and IPv6 in wireless networks

**Week 13: Revision**

- Comprehensive review of topics covered
- Practice exercises and quizzes

**Week 14: Examinations**

- Final assessments and examinations
- Evaluation of understanding and application of course material

**Note:** The course outline can be adjusted based on the specific needs and depth of coverage desired for each topic. Additionally, practical sessions, case studies, and hands-on projects can be incorporated throughout the course to enhance practical skills and understanding.

**Teaching Methodologies**

Lectures, practical and tutorial sessions in Computer Laboratory, individual and group assignments, exercises and project work Instructional Materials/Equipment Overhead projector and computer, handouts, white boards, appropriate software.

**Course Assessment**

- 30% Continuous Assessment (Tests 10%, Assignment 10%, Practical 10%)
- 70% End of Semester Examination.

**Core Textbooks**


1. Yi Bing Lin (2000). Wireless and Mobile Networks Architecture, ISBN-13: 978- 0471394921
2. Lee, Watson, Wugofski (2000). The Beginning WML and WML Script, ISBN-13: 978- 1861004581
3. Devi Kamal (2012). Mobile Computing, Oxford 2nd Edition, ISBN-13: 978- 0198068914